CCTV and Installation Fundamentals

AITS Learning Hub training manual and practice workbook

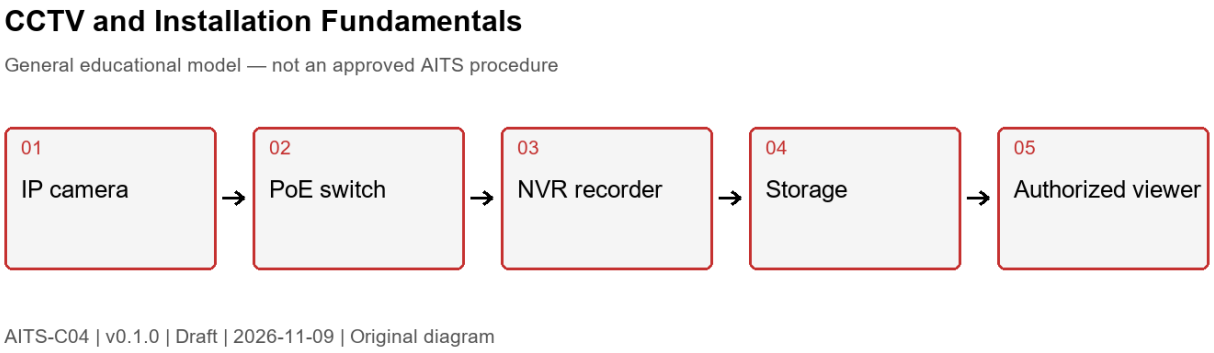
Sample Content — Pending Company Approval

This course introduces the concepts below and gives you exercises to practise them. Use the source notes to distinguish public company information, general guidance and fictional examples. It does not authorize live work or establish internal policy.

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Duration	About 100 minutes including exercises
Language	English; Thai summaries pending native-speaker review

Learning objectives

- Trace a basic CCTV signal path.
- Review power compatibility and cabling before connection.
- Prepare a survey that connects camera purpose to site constraints.
- Separate symptom diagnosis from authorized corrective work.



Cameras recorders and viewing

AITS-C04-L01 | 15 minutes | Draft

Objective: Trace a basic CCTV signal path.

Prerequisites: AITS-C01

A CCTV system combines image capture, transport, recording or storage, and authorized viewing. IP cameras send digital video over a network to an NVR or video-management system; analog cameras commonly send video to a DVR for recording. Hybrid designs exist, so verify the actual equipment rather than relying on the name alone.

A visible live image does not prove recording, time accuracy, playback or export works. A complete handover test considers the customer's agreed purpose and approved requirements. Camera resolution alone does not guarantee identification: lens, distance, angle, lighting and motion matter. All examples here are generic; AITS-specific acceptance criteria have not been supplied.

Worked example

Synthetic system: two IP cameras connect to a PoE switch; the recorder and authorized viewing workstation share the permitted network.

Try it

Draw the path from a camera to stored footage and identify where recording could fail while live viewing still works.

Your response: _____

Compare your reasoning

Camera → switch/network → recorder → storage. A disk or recording-schedule problem can affect storage while live video remains visible.

Remember

Capture, transport, record and view are distinct NVR and DVR describe common architectures Test playback as well as live view

Thai summary (AI translation, not professionally reviewed): แยกการรับภาพ การส่งข้อมูล การบันทึก และการดูภาพ ตรวจสอบการเล่นย้อนหลังด้วย ไม่ใช่ดูภาพสดอย่างเดียว

References: NET, EX. See source register.

PoE cabling and equipment checks

AITS-C04-L02 | 15 minutes | Draft

Objective: Review power compatibility and cabling before connection.

Prerequisites: AITS-C04-L01

Power over Ethernet can deliver power and data over compatible Ethernet cabling. Check both the device's supported PoE type and the switch or injector's capability. Do not assume every RJ45 port supplies power or every injector is standards-compatible. Manufacturer documentation takes priority for the actual model.

A switch has per-port limits and a total power budget. Compare planned loads against the documented budget, allowing for the manufacturer's conditions and cable losses. Axis explains that typical and maximum camera consumption differ with features and operating conditions. Inspect cable category, route, terminations, labeling, damage and suitability for the environment. Never improvise mains wiring as part of a network exercise.

Worked example

Four fictional cameras each require a documented maximum allocation of 15 W. Their total is 60 W; a 50 W usable budget is insufficient even before other loads.

Try it

Calculate the example load and list two checks beyond total watts.

Your response: _____

Compare your reasoning

60 W exceeds 50 W. Also verify per-port support and the correct PoE type for every device.

Remember

Check PoE type and per-port limit Check total usable power budget Inspect and label the cable path

Thai summary (AI translation, not professionally reviewed): ตรวจสอบชนิด PoE กำลังต่อพอร์ต และกำลังรวมของสวิตช์ กำลังใช้งานทั่วไปอาจต่ำกว่ากำลังสูงสุด

References: POE, EX. See source register.

Survey placement and installation preparation

AITS-C04-L03 | 15 minutes | Draft

Objective: Prepare a survey that connects camera purpose to site constraints.

Prerequisites: AITS-C04-L02

Start with the intended task: observe an area, recognize activity or capture usable identification detail under agreed conditions. Record field of view, expected lighting, mounting constraints, cable route, weather exposure and maintenance access. Avoid assumptions about recording private areas or neighboring property; escalate privacy and lawful-purpose questions to the responsible reviewer.

Before installation, check model and quantities against the approved scope, inspect for damage and confirm compatible accessories. Verify that safe access and required permissions are ready. A drawing should label devices and cable routes clearly enough for another technician to follow. Do not treat a draft survey as permission to drill, energize or change a customer system.

Worked example

A fictional entrance faces bright afternoon sunlight. A site review should test the relevant lighting conditions instead of accepting a clear indoor bench image.

Try it

Create a survey checklist with purpose, scene, route, power, privacy and access headings.

Your response: _____

Compare your reasoning

For each heading record known evidence, an unresolved question and the person who must confirm it; leave unknown measurements blank.

Remember

Define the viewing task Check real site conditions Confirm scope and safe access

Thai summary (AI translation, not professionally reviewed): สำรวจตามวัตถุประสงค์ของภาพ ตรวจแสง เส้นทางสาย ความเป็นส่วนตัว และทางเข้าทำงานที่ปลอดภัยก่อนติดตั้ง

References: EX, HEIGHT, QA. See source register.

Faults maintenance and handover

AITS-C04-L04 | 15 minutes | Draft

Objective: Separate symptom diagnosis from authorized corrective work.

Prerequisites: AITS-C04-L03

For a missing image, establish whether one camera or many are affected, when it began and whether live view or playback is missing. Observe authorized power/link indications and compare with a known-good path where permitted. Possible causes include power, cabling, addressing, authentication, stream compatibility and recording settings; do not jump straight to a factory reset.

Maintenance starts with approved intervals and model instructions. Review authorized health indicators, time accuracy, storage condition and image obstruction without changing settings unnecessarily. Handover evidence should identify devices, configuration versions, tests performed, results, limitations and open issues. Transfer credentials through the approved secure method, never in the ordinary completion report.

Worked example

Only playback is missing for camera C2; live view works. Investigate recording schedule, storage and timestamps before replacing the camera.

Try it

Write a handover entry that records a passed live-view test and an unresolved playback problem.

Your response: _____

Compare your reasoning

C2 live view passed at 14:00 in the agreed scene. Playback unavailable for the requested period. Recording/storage review pending; acceptance not claimed.

Remember

Scope the symptom before changing equipment Avoid destructive resets Record results and unresolved items

Thai summary (AI translation, not professionally reviewed): เริ่มจากขอบเขตอาการ ตรวจสอบอย่างมีเหตุผลและบันทึกผลกับงานค้าง ห้ามรีเซ็ตหรือเปิดเผยรหัสผ่านโดยไม่มีการอนุญาต

References: EX, QA, POE. See source register.

Practical scenarios

Insufficient PoE budget

Synthetic classroom exercise; not an AITS procedure. Role: Installation planner

Four fictional cameras need 15 W each; the usable switch budget is 50 W. Per-port compatibility is not yet checked.

Decide: What would you check first, and why? What requires confirmation or authorization? What would you record and hand over?

Appropriate: Flag the 60 W total and verify a suitable approved design.

Inappropriate: Approve because four ports are available.

Consequence: An undersized budget can produce unstable operation.

Poor entrance image

Synthetic classroom exercise; not an AITS procedure. Role: Technician trainee

A camera looks clear on the bench but silhouettes visitors at the entrance. Afternoon backlight differs from bench conditions.

Decide: What would you check first, and why? What requires confirmation or authorization? What would you record and hand over?

Appropriate: Review scene, lens and lighting against the agreed purpose.

Inappropriate: Claim megapixels alone guarantee identification.

Consequence: Testing relevant conditions reveals whether the objective is met.

Live view without playback

Synthetic classroom exercise; not an AITS procedure. Role: Support trainee

Fictional C2 has live video but no playback for yesterday. Other camera recordings exist; no reset is authorized.

Decide: What would you check first, and why? What requires confirmation or authorization? What would you record and hand over?

Appropriate: Check recording path and preserve evidence.

Inappropriate: Reset every camera.

Consequence: Targeted diagnosis avoids unnecessary disruption.

Use the full scenario JSON for the four-criterion rubric. Score evidence, decision quality, authority/safety and handover from 0–2 each. 0 = absent or unsafe; 1 = partial; 2 = complete and supported. Scores are formative, not certification.

Knowledge check

1. Which device commonly records network-camera streams?

A. NVR | B. Passive patch panel | C. Unpowered cable reel

2. Does live viewing prove recording is working?

A. Yes | B. Only in daytime | C. No

3. What influences useful image detail?

A. Only the brand sticker | B. Lens, distance, lighting and motion | C. Only the camera color

4. Can four 15 W allocations fit a 50 W usable budget?

A. Yes, total is 30 W | B. Yes, port count is enough | C. No, total is 60 W

5. Does every RJ45 port provide PoE?

A. Only when a camera is connected | B. No | C. Yes

6. Why check maximum rather than only typical consumption?

A. Operating features can raise demand | B. Typical always means maximum | C. It changes the camera IP address

Use the answer key after attempting the questions. Proposed practice pass mark: 80%. This is not an AITS competency rule.

Knowledge check continued

7. What should guide camera placement first?

A. A random mounting point | B. The agreed viewing purpose and scene | C. The shortest cable at any cost

8. What should happen before drilling an unfamiliar wall?

A. Verify permission, hidden-service risks and approved method | B. Drill a test hole | C. Ask the AI to guess

9. Is a draft survey permission to install?

A. Yes | B. Only for small cameras | C. No

10. Live view works but playback fails. Which area deserves review?

A. Recording settings and storage | B. Camera paint color | C. Only the monitor stand

11. What is a poor first response to a missing image?

A. Determine affected devices | B. Record the start time | C. Factory reset without authorization

12. Customer passwords belong in the ordinary handover report.

A. True | B. False

Use the answer key after attempting the questions. Proposed practice pass mark: 80%. This is not an AITS competency rule.

Answer key and references

1. A — An NVR receives network video for recording.

2. C — Storage and recording settings need separate verification.

3. B — Image usefulness depends on the scene and imaging conditions.

4. C — Power budget and port count are separate constraints.

5. B — PoE capability must be verified.

6. A — Load can change with IR, heaters and other functions.

7. B — Placement must support the intended task.

8. A — Site hazards and authorization require verification.

9. C — Planning evidence is distinct from authorization.

10. A — The evidence points toward the recording path, though diagnosis remains open.

11. C — A reset can destroy configuration and evidence.

12. B — Routine reports should not expose secrets.

EX: Original instructional examples. 2026-11-09. training-content/build_content.py

HEIGHT: ILO working at height. Accessed 2026-11-09. https://www.ilo.org/topics-and-sectors/occupational-safety-and-health-guide-labour-inspectors-and-other/working-height

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